TCS Numerical Ability Preparation One Shot | Aptitude Prep for TCS NQT & TCS Ignite Smart ...

$$\text{Speed} = \frac{\text{distance}}{\text{time}}$$

time constant

if distance constant,

$$\text{speed} \propto \frac{1}{\text{time}}$$

Speed ratio ⇒ a : b

time ratio ⇒ b : a

Same distance

A = 2 kmph → 3 hrs

B = 3 kmph → ? hrs

Speed ratio

A : B

2 : 3

Time ratio

A : B

3 : 2

3 hrs 2 hrs

6:16:30 / 11:50:56 TCS Speed, Time & Distance

Speed Time & Distance

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$$\boxed{18 \text{ km/hr} = 5 \text{ m/s}}$$

$$1 \text{ km/hr} = \frac{5}{18} \text{ m/s}$$

$$\frac{18}{5} \text{ km/hr} = 1 \text{ m/s}$$

$18 + 18 + 9 = 45$

45 km/hr

18 km/hr 18 km/hr 9 km/hr

5 m/s 5 m/s 2.5 m/s

12.5 m/s

45 km/hr → — m/s

18 km/hr → 5 m/s

1 km/hr → $\frac{5}{18}$ m/s

$$45 \text{ km/hr} = \frac{5}{18} \times 45 \text{ m/s}$$

$$= \frac{5}{\cancel{18}_{2}} \times \cancel{45}^{5} = \frac{25}{2}$$

$$= 12.5$$

Speed Time & Distance

6:21:29 / 11:50:56 TCS Speed, Time & Distance >

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Q1. Tesla Car travelling at the rate of 45 kmph. How many seconds, it will take to cover a distance of 4/5 km?

(M1) $S = \frac{d}{t} \Rightarrow t = \frac{d}{S} = \frac{\frac{4}{5}}{45} = \frac{4}{5 \times 45}$ hrs

$= \frac{4}{5 \times 45} \times 60 \times 60$ seconds

$= \frac{4}{\cancel{5} \times \cancel{45}_{\cancel{9}}} \times \cancel{60}^{4} \times \cancel{60}^{\cancel{20}\,4}$

$= 4 \times 4 \times 4 = 64$ seconds

1 hr = 60 mins

1 min = 60 seconds

60 min = 60×60

1 hr = 60 × 60 seconds

$\frac{4}{5 \times 45}$ hr $= \frac{4}{5 \times 45} \times 60 \times 60$

Q2. A person is driving a car at a speed of 40 kmph can complete a journey in 9 hours. How long will it take to travel the same distance at 60 kmph?

M1

$s = \frac{d}{t} \Rightarrow d = s \times t$

$= 40 \times 9$

$= 360$ KM

$s = \frac{d}{t} \Rightarrow t = \frac{d}{s} = \frac{360}{60} = 6$ hrs

M2 distance same

Speed ratio = 40 : 60

= 4 : 6

= 2 : 3

Time ratio = 3 : 2 (×3, ×3) → 9 : 6

9 : 6 = 3 : 2

Ans: 6 hrs

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CONCEPT 2 – AVERAGE SPEED

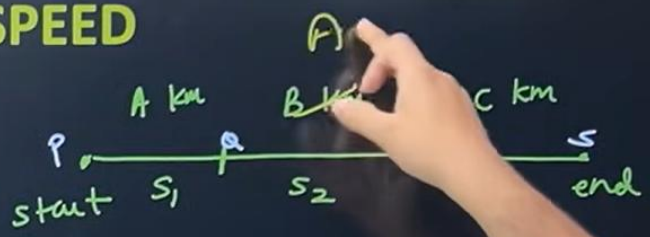
$$\text{Speed} = \frac{\text{Distance}}{\text{Time}}$$

$$S = \frac{d}{t}$$

$$t = \frac{d}{S}$$

$$\text{Average Speed} = \frac{\text{Total Distance}}{\text{Total Time}}$$

$$= \frac{A+B+C}{\frac{A}{S_1} + \frac{B}{S_2} + \frac{C}{S_3}}$$



$$\text{Avg Speed} = \frac{S_1 + S_2}{3}$$

PQ $\quad t_1 = \frac{A}{S_1}$

QR $\quad t_2 = \frac{B}{S_2}$

RS $\quad t_3 = \frac{C}{S_3}$

Speed Time & Distance

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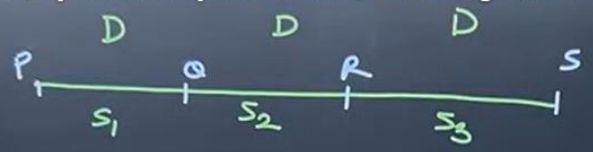
Distance same

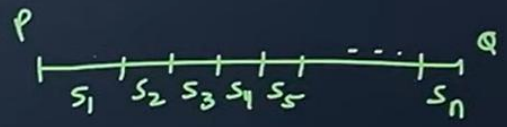


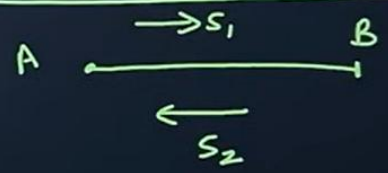
$$\text{Avg Speed} = \frac{D+D+D}{\frac{D}{S_1}+\frac{D}{S_2}+\frac{D}{S_3}}$$

$$= \frac{3D}{D\left[\frac{1}{S_1}+\frac{1}{S_2}+\frac{1}{S_3}\right]}$$

$$= \frac{3}{\frac{1}{S_1}+\frac{1}{S_2}+\frac{1}{S_3}}$$

$$\text{Avg Speed} = \frac{n}{\frac{1}{S_1}+\frac{1}{S_2}+\frac{1}{S_3}+\dots\frac{1}{S_n}}$$

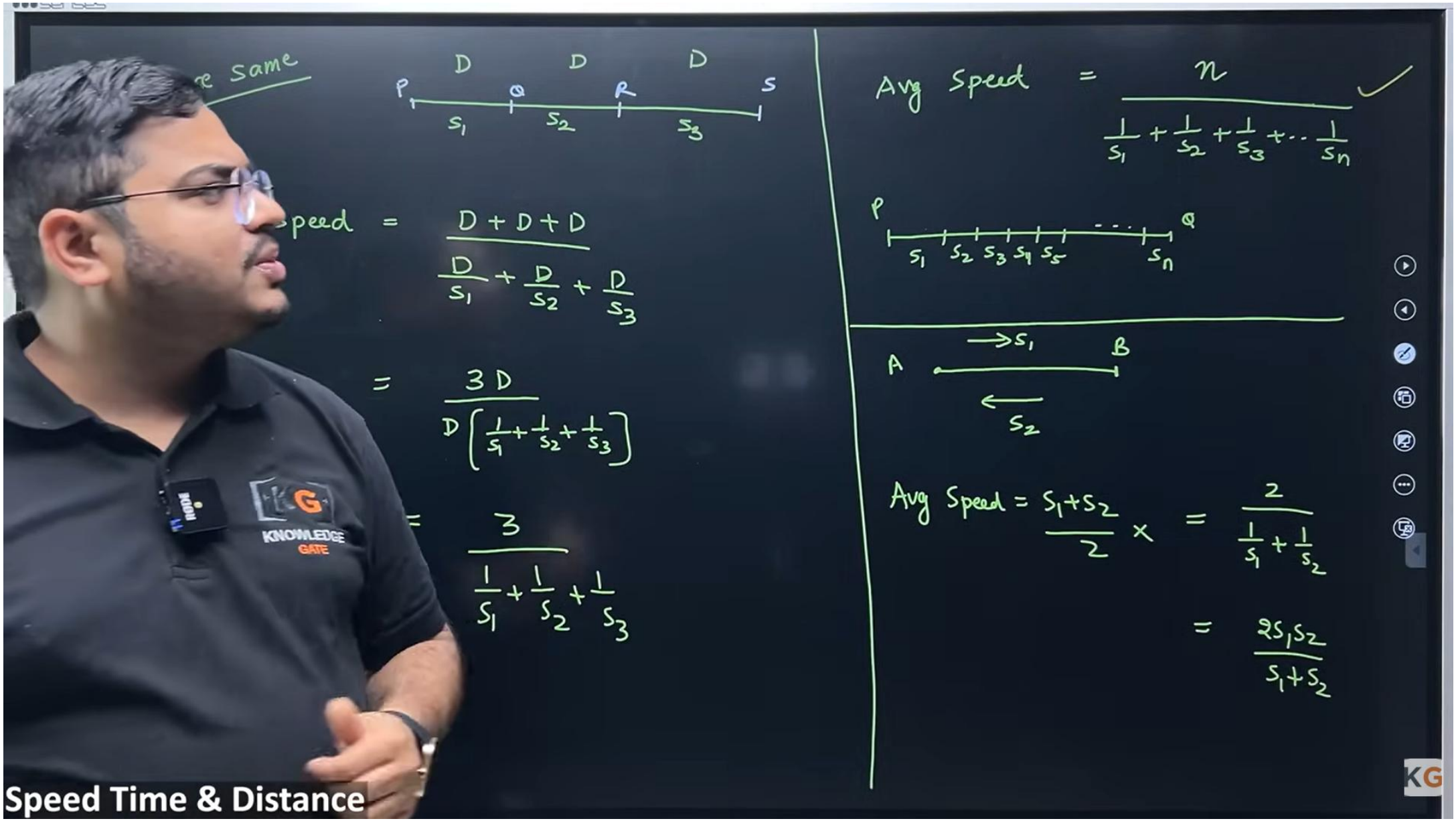


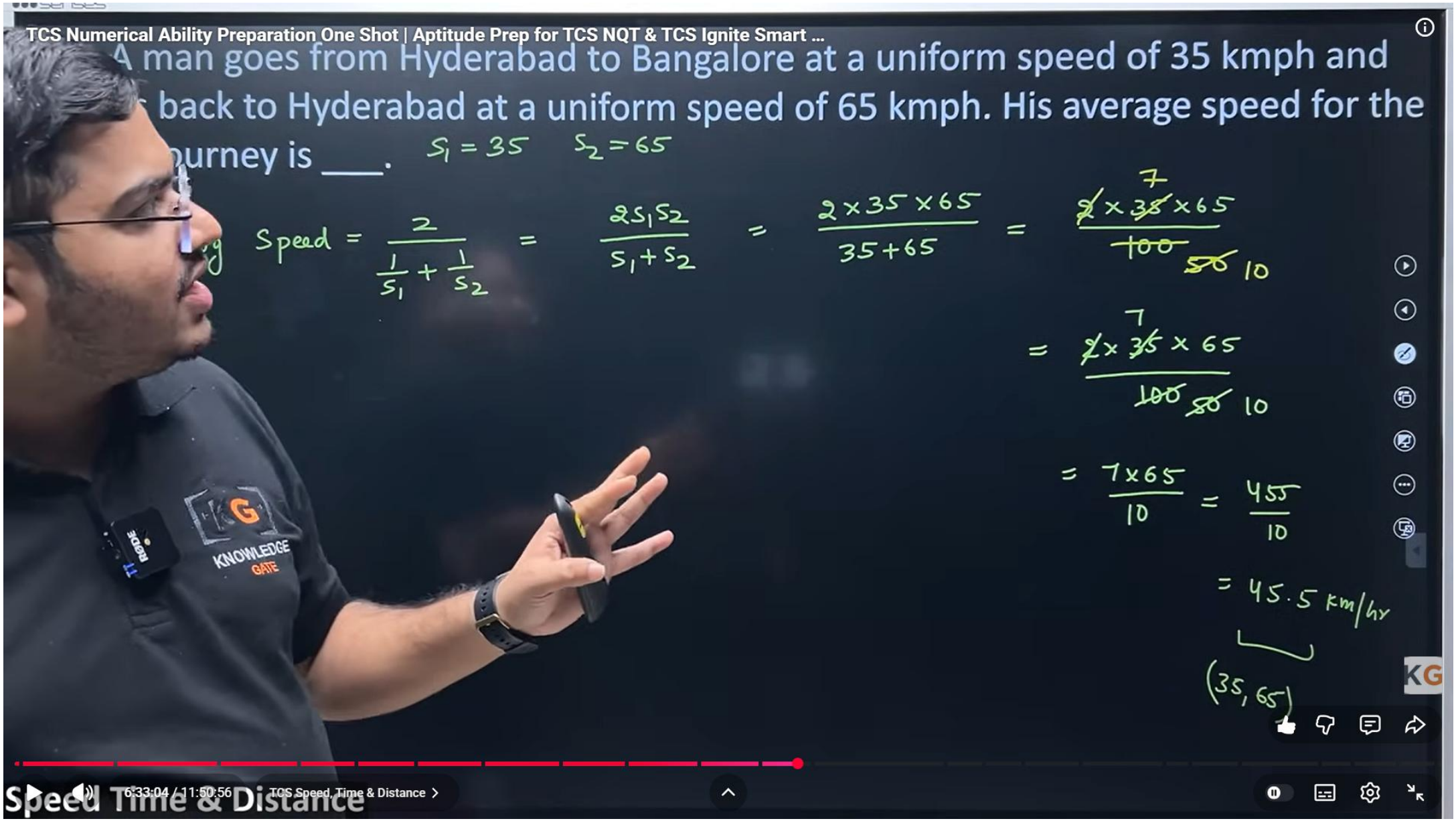


$$\text{Avg Speed} = \frac{S_1+S_2}{2} \times = \frac{2}{\frac{1}{S_1}+\frac{1}{S_2}}$$

$$= \frac{2S_1S_2}{S_1+S_2}$$

KG KNOWLEDGE GATE


e same
D D D
P Q R S
s_1 s_2 s_3
peed = $\frac{D+D+D}{\frac{D}{s_1}+\frac{D}{s_2}+\frac{D}{s_3}}$
$= \frac{3D}{D\left[\frac{1}{s_1}+\frac{1}{s_2}+\frac{1}{s_3}\right]}$
$= \frac{3}{\frac{1}{s_1}+\frac{1}{s_2}+\frac{1}{s_3}}$
Avg Speed = $\frac{n}{\frac{1}{s_1}+\frac{1}{s_2}+\frac{1}{s_3}+\dots\frac{1}{s_n}}$
P Q
s_1 s_2 s_3 s_4 s_5 s_n
A B
s_1
s_2
Avg Speed = $\frac{s_1+s_2}{2}$ × = $\frac{2}{\frac{1}{s_1}+\frac{1}{s_2}}$
$= \frac{2s_1s_2}{s_1+s_2}$
KNOWLEDGE GATE
Speed Time & Distance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A man goes from Hyderabad to Bangalore at a uniform speed of 35 kmph and back to Hyderabad at a uniform speed of 65 kmph. His average speed for the journey is ___.
S1 = 35 S2 = 65
Speed = 2 / (1/S1 + 1/S2) = 2S1S2 / (S1+S2) = 2×35×65 / (35+65) = 2×35×65 / 100
= 2×35×65 / 100
= 7×65 / 10 = 455 / 10
= 45.5 km/hr
(35, 65)
KNOWLEDGE GATE
6:33:04 / 11:50:56
TCS Speed, Time & Distance
Speed Time & Distance

Q4. Peter travels from his home to a famous tea stall. One-third of his journey is covered at the rate of 25 kmph, one-fourth at the rate of 30 kmph and the rest at 50 kmph. The average speed for the whole journey is __

$s = \frac{d}{t}$

$t = \frac{d}{s}$

M2 $\frac{1}{3}, \frac{1}{4}$ LCM(3,4) = 12

Assume, total distance = 12 km

$\frac{1}{3}^{rd} \rightarrow \frac{1}{3} \times 12 = 4$ km

$\frac{1}{4}^{th} \rightarrow \frac{1}{4} \times 12 = 3$ km

remaining = 5 km

12 km

4 km	3 km	5 km
25	30	50
$t = \frac{4}{25}$	$\frac{3}{30}$	$\frac{5}{50}$
$\frac{4}{25}$	$\frac{1}{10}$	$\frac{1}{10}$

total time $= \frac{4}{25} + \frac{1}{10} + \frac{1}{10} = \frac{4}{25} + \frac{2}{10} = \frac{4}{25} + \frac{1}{5}$

$= \frac{4}{25} + \frac{5}{25} = \frac{9}{25}$